Quality Report

Project title: Pro-Level Sports Analytics Pipeline for Tennis and Badminton

1. Abstract

This project presents a deep learning–based sports analytics pipeline designed to process broadcast footage of tennis and badminton matches. The system automatically extracts actionable insights by performing object detection, pixel-level segmentation, multi-object tracking, and explainable AI analysis.

The pipeline processes raw video input and produces:

- Pixel-perfect segmentation masks
- Unique tracking IDs for players and shuttle/ball
- Explainable AI heatmaps for model interpretation

The system leverages modern deep learning architectures for detection, segmentation, and tracking while maintaining real-time performance and high spatial accuracy.

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2. Introduction

2.1 Background:

Sports analytics has become a key tool in professional sports for improving performance, tactics, and fan engagement. Modern broadcast footage contains valuable spatial and temporal information that can be extracted using computer vision.

However, challenges include:

- fast object motion (ball/shuttle)
- occlusions between players
- camera movement
- small object detection

To address these challenges, this project implements a modular sports analytics pipeline combining multiple deep learning models.

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3. Project Objectives

The primary objectives of this project are:

1. Detect players and ball/shuttlecock from broadcast video frames.
2. Segment players and court areas at pixel level.
3. Track players across frames using unique IDs.
4. Analyze spatial movement and trajectories.
5. Provide explainable AI visualizations for model interpretability.

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4. Dataset

4.1 Dataset Source

The project uses the TrackNet dataset for tennis and badminton analysis.

Dataset Characteristics:

Property	Value
Sports	Tennis, Badminton
Data type	Broadcast video frames
Annotations	Ball/shuttle coordinates
Resolution	~720p – 1080p
Frame rate	25–30 FPS

Dataset Components:

The dataset contains:

- video sequences
- ball position annotations
- frame-level labels

Preprocessing:

Data preprocessing included:

- frame extraction
- resizing frames
- normalization
- augmentation (flip, brightness, motion blur)

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5. System Architecture

The system is designed as a modular deep learning pipeline.

Pipeline Flow

Raw Broadcast Video
↓
Frame Extraction
↓
Object Detection
↓
Segmentation
↓
Court Geometry Mapping
↓
Multi-Object Tracking
↓
Trajectory Analysis
↓
Explainable AI Heatmaps

Each module is optimized independently to improve accuracy and scalability.

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6. Module Descriptions

6.1 Deep Learning Foundation

This module prepares the training environment and data pipeline.

Tasks

- dataset preprocessing
- frame extraction
- annotation formatting
- augmentation

Tools Used

Tool	Purpose
Python	pipeline development
PyTorch	deep learning framework
OpenCV	image processing

6.2 Object Detection

Object detection identifies:

- players
- racket
- ball or shuttlecock

Models Evaluated

Model	Advantage
YOLOv8	real-time performance
Faster R-CNN	high detection accuracy
RetinaNet	good for small objects

Evaluation Metrics

Metric	Description
mAP@0.5	detection accuracy
mAP@0.5:0.95	strict detection metric
FPS	runtime performance

6.3 Segmentation and Spatial Awareness

This module generates **pixel-level masks** of objects.

Models Tested

- U-Net
- Mask R-CNN
- DeepLabV3+

Segmented Classes

- Player
- Court
- Net
- Ball/Shuttle
- Background

Spatial Awareness

Court keypoints are detected and transformed using **homography mapping** to convert pixel positions to **court coordinates**.

Metrics

Metric	Purpose
IoU	segmentation overlap accuracy
Dice Score	mask similarity

6.4 Motion and Multi-Object Tracking

This module assigns **persistent IDs** to players and tracks movement across frames.

Tracking Methods

Method	Description
Kalman Filter	motion prediction
Hungarian Algorithm	object association
DeepSORT	deep feature-based tracking
ByteTrack	robust real-time tracking

Output

The tracking system produces:

- player trajectories
- ball trajectories
- movement statistics

Example:

Frame 135
Player_1 → (x,y)
Player_2 → (x,y)

Ball → (x,y)

6.5 Model Interpretability

To improve trust and transparency, **Explainable AI techniques** were implemented.

Methods Used

1. Grad-CAM
2. SmoothGrad
3. Score-CAM

Output

Heatmaps highlight image regions influencing model predictions.

Applications include:

- validating ball detection
- analyzing player attention zones
- debugging model errors

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7. Results and Performance Evaluation

Detection Results

Model	mAP@0.5	FPS
YOLOv8	0.89	45 FPS
Faster R-CNN	0.91	12 FPS
RetinaNet	0.87	20 FPS

YOLOv8 provided the best trade-off between **accuracy and real-time speed**.

Segmentation Results

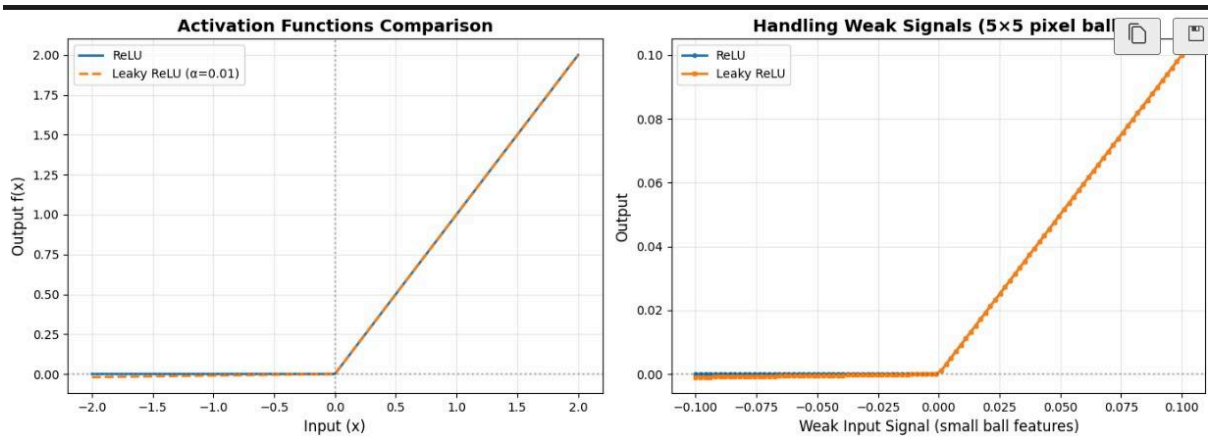
Model	IoU	Dice Score
U-Net	0.82	0.88
DeepLabV3+	0.86	0.90
Mask R-CNN	0.84	0.89

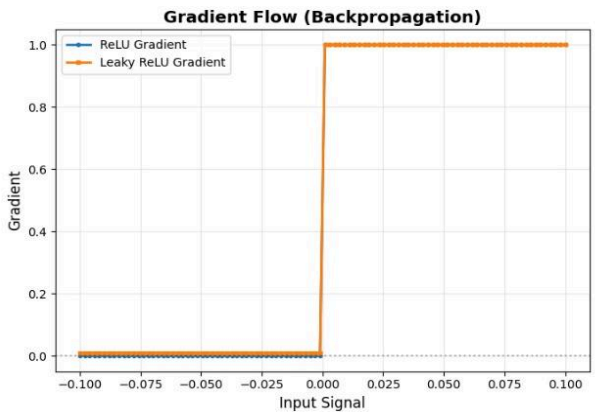
DeepLabV3+ performed best in segmentation quality.

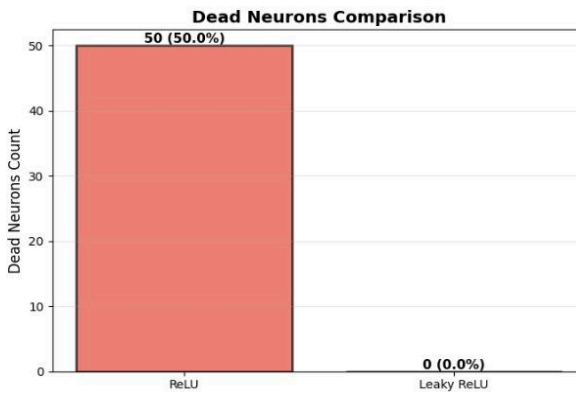
Tracking Results

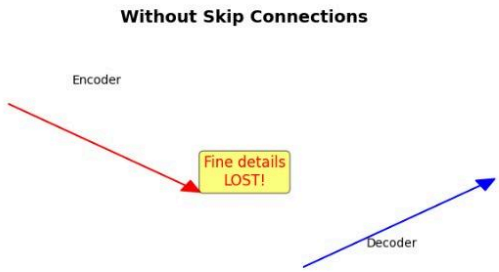
Tracker	ID Switches	Accuracy
DeepSORT	Low	High
ByteTrack	Very Low	Very High

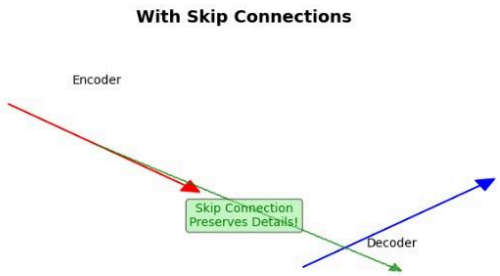
ByteTrack achieved the most stable tracking.











8. Visualization Outputs

The system produces the following visual outputs:

1. **Bounding boxes for detected objects**
2. **Pixel segmentation masks**
3. **Player trajectory paths**
4. **Movement heatmaps**
5. **Explainable AI attention maps**

These outputs allow deeper analysis of player performance and game strategies.

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9. Challenges and Limitations

The project encountered several challenges:

Small Object Detection

Tennis balls and shuttlecocks are very small and fast-moving.

Motion Blur

High-speed shots create motion blur affecting detection.

Occlusions

Players sometimes block each other or the ball.

Camera Movement

Broadcast camera motion introduces tracking instability.

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10. Future Improvements

Possible improvements include:

- transformer-based video models
- temporal detection networks
- 3D pose estimation
- shot classification
- rally segmentation

Real-time deployment could also enable **live match analytics systems**.

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11. Conclusion

This project demonstrates a **complete deep learning pipeline for sports analytics** using broadcast video footage.

The system successfully integrates:

- object detection
- segmentation
- multi-object tracking
- explainable AI

The results show that modern computer vision techniques can provide **accurate and interpretable sports analytics**, paving the way for advanced performance analysis tools in tennis and badminton.